

HW_04

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Due: 5:00pm, 29 September

Create a self-contained R script or Rmarkdown file to do the following:
Simulate independent double observer data with the following properties:

- $n_{\text{Sites}}=200$
- $\lambda=3$
- $p_1=0.3$ and $p_2=0.5$

```
set.seed(314159)
```

```
nSites <- 200
```

```
lambda <- 3
```

```
N <- rpois(nSites, lambda)
```

```
summary(N)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00   2.00   3.00   2.91   4.00   8.00
```

```
mean(N)
```

```
## [1] 2.91
```

```
var(N)
```

```
## [1] 2.916482
```

```
p1 <- 0.3
```

```
p2 <- 0.5
```

```
pi1 <- p1*(1-p2)
```

```
pi2 <- (1-p1)*p2
```

```
pi3 <- p1*p2
```

```
pi4 <- (1-p1)*(1-p2)
```

```
cp <- c(pi1,
        pi2,
```

```

      pi3
    )

y <- matrix(NA,
            nrow=nSites,
            ncol=3
)

colnames(y) <- c("obs1_only",
                 "obs2_only",
                 "both"
)

for(i in 1:nSites) {
  counts <- rmultinom(1, size=N[i], prob = c(cp,pi4))
  y[i,] <- counts[1:3]
}

head(y)

```

```

##      obs1_only obs2_only both
## [1,]         0         1    0
## [2,]         2         0    0
## [3,]         1         1    0
## [4,]         0         1    0
## [5,]         0         4    0
## [6,]         0         0    0

```

```
summary(y)
```

```

##      obs1_only      obs2_only      both
## Min.   :0.000   Min.   :0.0   Min.   :0.000
## 1st Qu.:0.000   1st Qu.:0.0   1st Qu.:0.000
## Median :0.000   Median :1.0   Median :0.000
## Mean   :0.535   Mean   :0.9   Mean   :0.465
## 3rd Qu.:1.000   3rd Qu.:2.0   3rd Qu.:1.000
## Max.   :3.000   Max.   :4.0   Max.   :3.000

```

Fit the model using `unmarked`. You will need to create an observation covariate indicating observer A and B. When you create the `unmarkedFrameMPois`, use `type=double`.

```

observer <- matrix(
  c("A", "B"),
  nrow = nSites,
  ncol = 2,
  byrow = TRUE
)
colnames(observer) <- c(
  "observer1",
  "observer2"
)

```

```
library(unmarked)
umf <- unmarkedFrameMPois(
  y = y,
  obsCovs = list(observer = observer),
  type = "double"
)
```

Warning: obsCovs contains characters. Converting them to factors.

```
fm <- multinomPois(
  ~ observer -1 ~ 1,
  data = umf
)

fm
```

```
##
## Call:
## multinomPois(formula = ~observer - 1 ~ 1, data = umf)
##
## Abundance (log-scale):
## Estimate      SE      z P(>|z|)
##      1.08 0.0741 14.5 8.36e-48
##
## Detection (logit-scale):
##           Estimate      SE      z P(>|z|)
## observerA    -0.66 0.128 -5.171 2.33e-07
## observerB    -0.14 0.142 -0.989 3.23e-01
##
## AIC: 1254.228
## Number of sites: 200
```

```
unmarked_coef <- data.frame(
  lambda_hat = exp(coef(fm)["lambda(Int)"]),
  pA_hat = plogis(coef(fm)["p(observerA)"]),
  pB_hat = plogis(coef(fm)["p(observerB)"])
)

table(unmarked_coef)
```

```
## , , pB_hat = 0.465000112804334
##
##           pA_hat
## lambda_hat 0.34065945180332
## 2.9354835201373 1
```

Fit the same model in JAGS

For both unmarked and JAGS, use the point estimates of p_1 and p_2 to compute the π probabilities.